A PROJECT REPORT ON

IMAGE SEGMENTATION: COMPARATIVE PERFORMANCE OF WATERSHED VS. GRAPH CUT FOR OBJECT ISOLATION

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CERTIFICATE OF AUTHENTICITY

This is to certify that the project report entitled “Image Segmentation: Comparative Performance of Watershed vs. Graph Cut for Object Isolation” is a bona fide record of the Computer Vision course case-design work carried out by Harsh Agarwal (Student Regd. No.: 23STUCHH010333), Aishwarya (Student Regd. No.: 23STUCHH010473), Shreshta (Student Regd. No.: 23STUCHH010484), Abhishek (Student Regd. No.: 24STUCHH010488), B.Tech – Artificial Intelligence & Machine learning , Section F, under the guidance of Dr. Kiran Kumar Eranki Department of CSE, Faculty of Science & Technology, IFHE University, Hyderabad.

The work presented in this report concerns the comparative implementation and visual evaluation of the Watershed and Graph Cut/GrabCut image-segmentation approaches using a natural mountain-lake image. The report has been prepared for academic evaluation as part of the Computer Vision course case-design activity.

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ABSTRACT

Image segmentation is a fundamental task of computer vision. Where an image is segmented into different regions or areas so that objects or parts of the scene can be separated from the surrounding background. This project will provide a comparative analysis of two traditional image segmentation methods, the Watershed algorithm. And Graph Cut segmentation, using GrabCut as an carry outation method via OpenCV. The images used to conduct the experiments were provided with a mountain-lake image for the experimental input and were processed using Python, OpenCV, NumPy, and Matplotlib under the Google Colab environment. The Watershed algorithm uses several preprocessing methods before applying the Watershed transformation: the input image is converted to grayscale, Gaussian smoothed, Otsu thresholded, morphological operations are performed, and a distance transform and connected components markers are created. The GrabCut method treats the image segmentation task as a labeling problem of separating the foreground from the background, using an initial rectangle to estimate the foreground and background color models. Both methods will produce intermediate segmentation masks and corresponding isolated-object outputs, which can be visually compared. The results from the supplied experiment indicate that the GrabCut method produces a larger connected region of the mountain-lake scene based on the selected image and initialization than the Watershed method; however, the Watershed method's foreground segmentation result is highly influenced by the thresholding and marker construction used. As a result, the study proves that the quality of the image segmentation is influenced by the algorithm used, the characteristics of the input image, the preprocessing steps performed, the quality of markers, and the initialization process. All comparisons between the two methods will be qualitative in nature due to the absence of a manually annotated ground-truth mask, thus no formal pixel-level accuracy, IoU, or F1 score will be claimed.

Keywords: Image Segmentation, Watershed, Graph Cut, GrabCut, Object Isolation, OpenCV, Computer Vision

PROJECT AIM: The objective of this project is to compare the Watershed and Grab Cut based image-semantic techniques and determine their relative efficacy in isolating objects or meaningful areas from an image.

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1. INTRODUCTION

1.1 Background and Context

Image segmentation feels like giving each pixel its identity based on how it looks or on numbers that describe it. Image segmentation is a step in computer vision because many tasks that come after need the picture to be broken into clear parts before any objects can be studied. Image segmentation is helpful for things like isolating objects, understanding scenes editing pictures analyzing images and automatic visual inspection.

The quality of image segmentation depends on how the picture's structure matches the assumptions of the chosen method. Natural scenes often have parts that differ in brightness, color, texture and edges. Therefore a method that works great on one image may need steps of preparation or different starting settings when applied to a different image.

Watershed treats an image like a landscape and uses the shape of regions and markers to find borders. I see that Graph Cut treats segmentation as a problem of labeling pixels as foreground or background and looks for a labeling that keeps data consistent while respecting the connection between neighboring pixels. In practice Graph Cut is carried out using OpenCV's GrabCut method.

1.2 PROBLEM STATEMENT

The main problem in this project is to separate objects or areas from a natural mountain-lake scene and to compare two segmentation methods under identical experimental settings. The input picture shows mountains, forest, water, reflections and sky. These parts produce borders and brightness patterns which make the picture a good test, for seeing how segmentation reacts to different preparation and starting choices.

The study looks at how Watershed and Graph Cut/GrabCut produce masks and separate-object images from the input. The test is mainly qualitative because the case here does not have a human-made ground-truth mask. So the results we see help us talk about how each method behaves, what it does well and what it struggles with in claiming a single best number.

1.3 Objectives of the Project

1. Load and preprocess the supplied natural RGB image using OpenCV.
2. Construct Watershed markers using thresholding, morphology, distance transformation and connected components.
3. Apply the Watershed transformation and generate an associated foreground mask.
4. Apply Graph Cut/GrabCut using an initial rectangle and generate a foreground mask.
5. Create isolated-object outputs from both segmentation masks.
6. Visually compare the segmentation outputs obtained from the two approaches.
7. Analyze the effect of thresholding, markers and initialization on segmentation quality.
8. Document the strengths, limitations, challenges and possible extensions of both methods.

1.4 Scope and Organization of the Report

The scope of the project is limited to comparing image segmentation techniques on a supplied natural-scene mountain-lake photo using computer vision methods. The work covers the process from loading the picture to cleaning it up segmenting it generating masks isolating objects and comparing the results visually. The report does not claim that the two methods are ranked for all cases because the experiment uses one image and has no ground-truth labels.

The report is organized as follows. Section 2 establishes the relevance of image segmentation and the comparative study, from academic and practical perspectives. Section 3 presents the methodology, including the input image, preprocessing pipeline, Watershed and Graph Cut algorithms, technology stack and implementation workflow. Section 4 presents the outputs and comparative discussion. Section 5 summarizes the achievements, recommendations, limitations and future scope. Section 6 lists the references while Section 7 provides the implementation code used in the experiment.

2. RELEVANCE OF THE PROJECT

2.1 Industrial and Computer Vision Significance

Image segmentation is a step in computer vision because it turns a complicated picture into parts that can be handled one by one. In computer vision pipelines separating the front part from the back part helps with analyzing objects measuring them recognizing them and making decisions later. Comparing ways of image segmentation matters when choosing the right method for a particular picture.

Watershed works well when clear borders can be shown with markers and the brightness pattern is clear. Graph Cut or GrabCut works well when a rough front part can be drawn and the front and back look different enough to guide the algorithm to label pixels step by step. This project shows these two image segmentation ideas on the same real scene so it is easier to see how they need to be set up and how they compare.

The project also shows that different image segmentation methods can react differently to the picture. Watershed mainly relies on brightness, how far things are, from each marker zone. GrabCut instead uses models of the back parts and the starting region.

Knowing these differences is key when building computer vision systems because a method works only when the algorithm fits the picture and the picture has the right traits.

2.2 Academic and Technical Alignment

From a perspective the project directly applies core Computer Vision concepts such as image segmentation, morphology, thresholding, distance transformation, connected components and graph-based foreground-background labeling. The project links ideas with a working implementation that uses Python and OpenCV.

The work also shows an end-to-end engineering process: defining a problem preprocessing image data building representations applying two algorithms producing visual outputs and interpreting limits. This makes the project design useful for testing understanding of classical computer vision techniques not just theoretical definitions.

Using Python and OpenCV gives technical alignment with tools that are commonly used for image-processing experiments. Implementing the methods of merely describing them lets the project show the link between algorithmic ideas, coding steps and visual results. This strengthens understanding of segmentation and gives a clear basis for comparing different approaches, under the same experimental conditions.

2.3 Practical Relevance of Comparative Segmentation

Segmentation algorithm cannot be selected independently of image characteristics. The way the method is initialized. Watershed pipeline in this project depends on the quality of the threshold-derived foreground and background regions. GrabCut depends on the rectangle and the color characteristics of the scene. Comparing Watershed and GrabCut on the input therefore highlights an important practical consideration: segmentation quality is a combined result of algorithm choice, preprocessing and parameter or initialization strategy.

The comparative approach helps identify the practical requirements of each method. Watershed requires markers and can be sensitive to incorrect region identification while GrabCut requires an appropriate initial foreground indication and may depend on the visual distinction between foreground and background. Observing these requirements on the image makes the differences between Watershed and GrabCut clearer and helps establish a practical understanding of segmentation workflow design.

Overall the project demonstrates that classical segmentation techniques remain useful for understanding computer vision processes. By implementing and comparing Watershed and GrabCut on the natural image the study provides a structured way to examine preprocessing, initialization, region formation and final segmentation outputs. The project therefore has relevance both as an exercise in computer vision and, as a practical demonstration of how segmentation methods are applied and evaluated.

3. METHODOLOGY

3.1 Dataset Profiling and Input Image Architecture

The experiment takes the mountain-lake photo that was provided as the starting image. The picture has a mountain range, areas with trees, a body of water reflections, stones and the sky. These elements create natural edges and sections that look different, from each other. The same photo is used in both segmentation processes so that the comparison is done using the same image.

Image Component	Observed Characteristics	Segmentation Relevance
Mountain region	Strong large-scale shape and boundary structure	Provides visible region boundaries for segmentation
Forest region	Dense texture with repeated tree structures	Can produce complex local regions and boundaries
Water region	Large relatively continuous region with reflections	Useful for examining connected foreground/background regions
Sky region	Broad region with different intensity and color	Contributes to background separation
Rocks/shore	Small high-contrast structures	May influence thresholding and marker formation

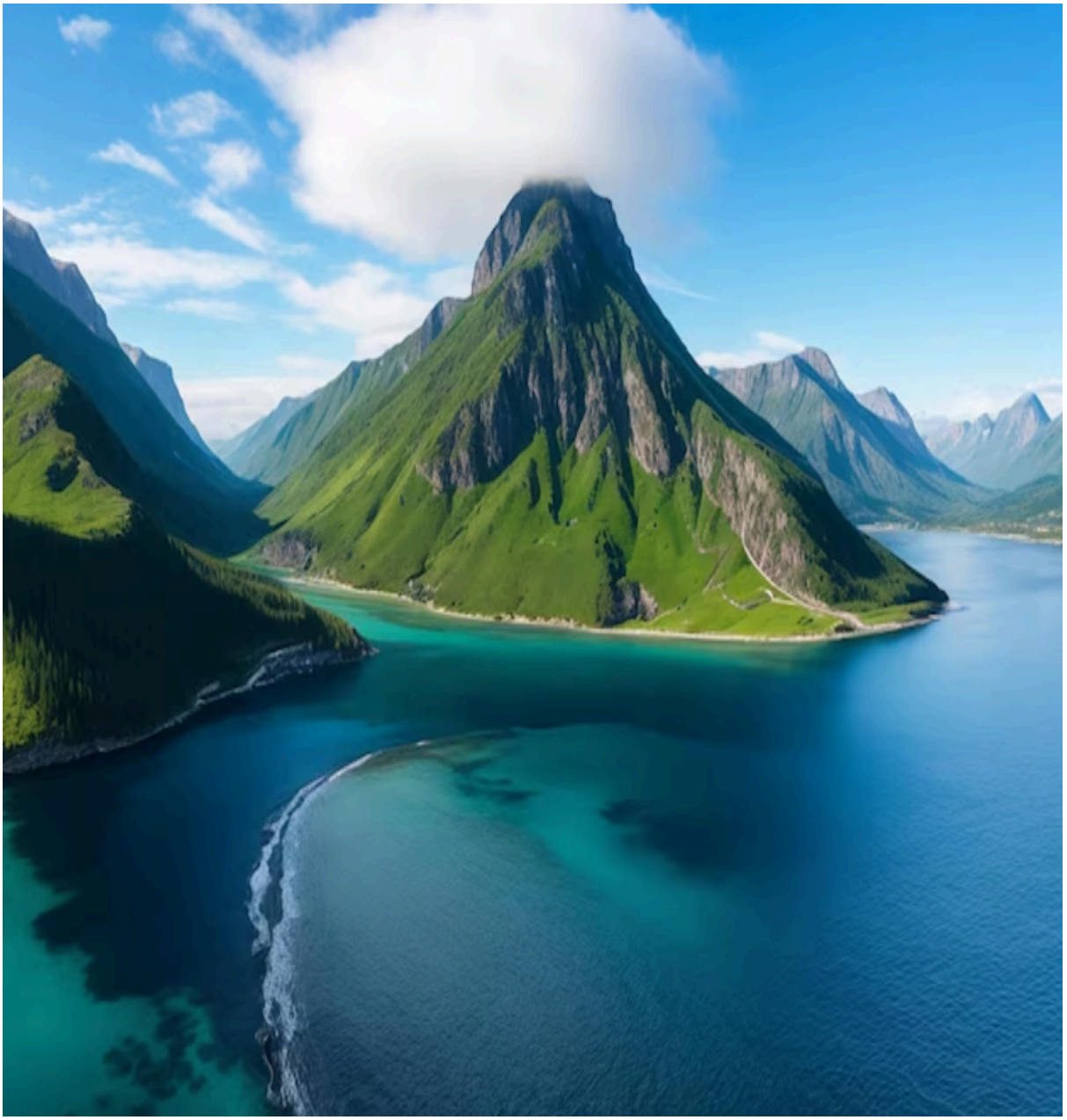


Figure 1: Supplied mountain-lake input image used for the segmentation experiment.

3.2 Image Preprocessing and Marker Engineering

The Watershed pipeline starts by taking the input image and changing it from BGR to grayscale. A Gaussian smoothing step is done to make changes in intensity less noticeable before the next step. Then Otsu thresholding is used to get an white version of the image. After that morphological opening is applied to get rid of unwanted parts. Dilation is then used to make the sure-background area bigger.

A distance transform is made from the cleaned-up foreground part. The pixels that are away from the background get higher distance values. Thresholding this distance map creates the sure-foreground area. Connected-component analysis is used next to make markers for the Watershed process. The part

that is not sure between the background and the foreground is considered unknown. Given the right marker value before the Watershed is run.

This part of the process is important because the final Watershed result depends directly on the markers, for the foreground and the background. If the thresholding step does not show the object the final segmentation might not be complete or might not look correct.

3.3 Applied Segmentation Algorithms

3.3.1 Watershed Segmentation

The Watershed method treats an image as a surface. In this view intensity structure helps to find boundaries between regions.In this implementation the marker-based pipeline first creates foreground and background information before the Watershed transformation. The resulting marker image contains information. The sure-foreground mask is also used to make an isolated-object image.

The practical strength of Watershed method is its ability to separate regions when meaningful markers and clear boundaries are available.The practical limitation of Watershed method in the demonstrated case is sensitivity, to thresholding and marker construction. Poor markers can cause an isolated-object mask to cover a small or incomplete portion of the intended scene.

3.3.2 Graph Cut / GrabCut Segmentation

Graph Cut segmentation is set up as a problem of labeling pixels as either foreground or background. In this method OpenCV’s GrabCut function starts with a rectangle that covers most of the image. This rectangle gives clues about which parts of the image are probably foreground and which are likely background.

GrabCut works by estimating color models for both the foreground and the background. It keeps improving the labels of each pixel based on these models. Then it combines the pixels that're definitely in the foreground with those that are possibly, in the foreground to create a final binary mask. This mask is used to separate the object from the rest of the image by applying it to the RGB image.

This approach works well when you can roughly show where the foreground is. But the result can be affected by the starting rectangle the nature of the image and how similar the foreground and background colors are.

3.4 Comprehensive Technology Stack and Implementation Workflow

Component	**Technology / Purpose**
Programming Language	Python
Computer Vision Library	OpenCV (cv2)
Numerical Processing	NumPy
Visualization	Matplotlib
Execution Platform	Google Colab
Input Format	JPG/PNG natural image
Segmentation Methods	Watershed and Graph Cut/GrabCut

The complete processing pipeline is: Upload Image → Read Image → Resize if Necessary → Watershed Preprocessing → Watershed Segmentation → Graph Cut Initialization → GrabCut Segmentation → Create Masks → Isolate Objects → Display and Save Results.

- Original Image: displays the selected natural scene before segmentation.
- Binary Image: shows the threshold-based representation used in Watershed marker preparation.
- Watershed Segmentation: displays the image with the Watershed boundary information.
- Watershed Object: displays the portion retained by the Watershed foreground mask.
- Graph Cut Mask: displays the foreground/background mask generated by GrabCut.
- Graph Cut Object: displays the image after applying the Graph Cut foreground mask.
- Final Segmentation: provides the final visual output used for comparison.

4. RESULTS AND DISCUSSION

4.1 Segmentation Output Evaluation

The Google Colab run generates a series of outputs both final. These outputs contain the image, a binary image, the Watershed segmentation the Watershed object, the Graph Cut mask, the Graph Cut object, an original comparison view and a final segmentation view. All of these outputs allow you to look at the masks and also at the intermediate representations that shape the segmentation.

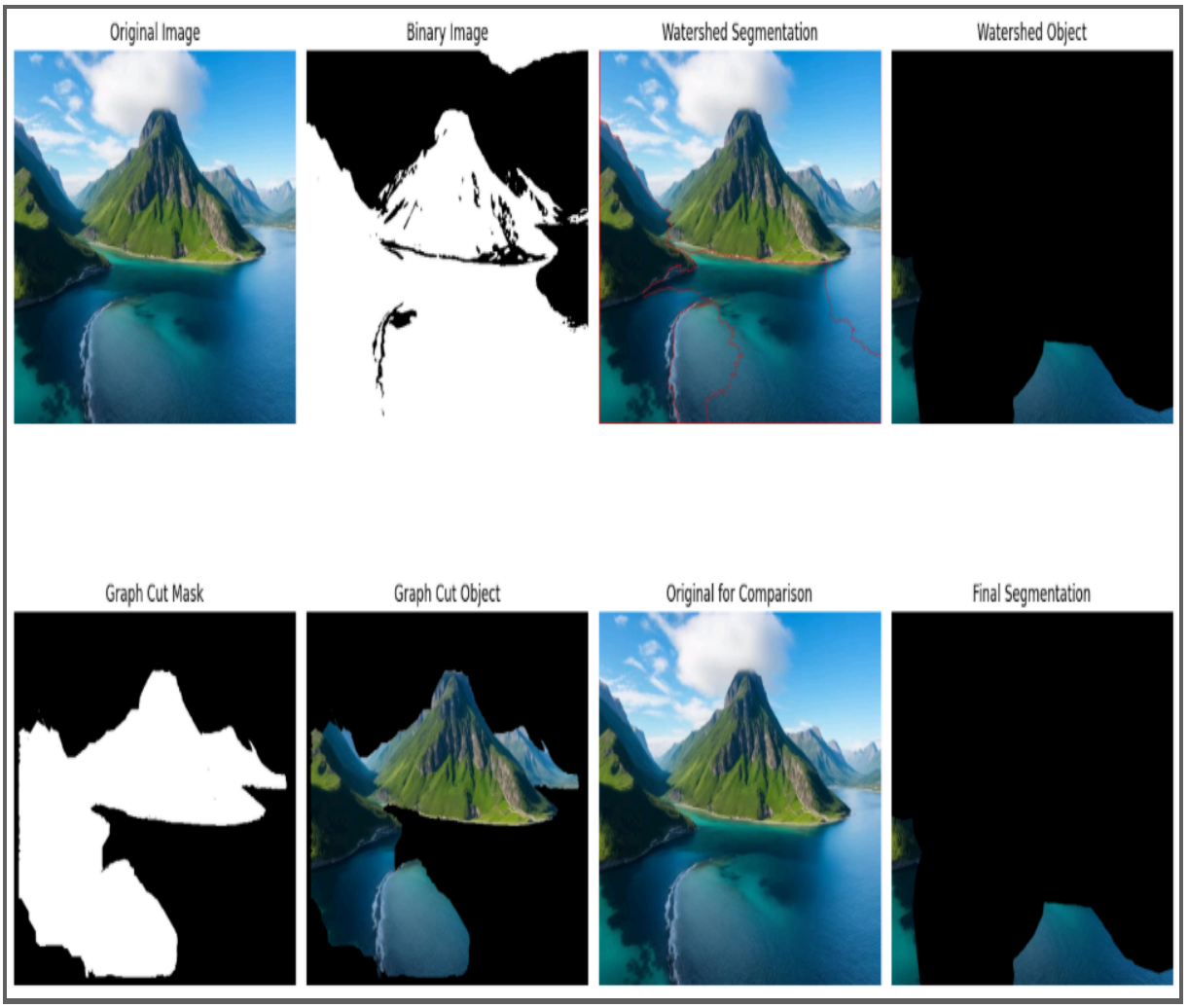


Figure 2: Supplied Google Colab output showing Watershed and Graph Cut/GrabCut processing stages.

The supplied report says that the Watershed pipeline creates a marker-based segmentation and the Watershed foreground mask keeps a small part of the picture. This shows that the chosen thresholding and markers do not match the foreground we want for this photo.

The report also says that the Graph Cut mask covers an area and the Graph Cut object keeps the mountain, lake and forest inside the foreground while hiding much of the background. With this image and starting point this is seen as the large-region isolation in the test.

These results only apply to this picture and how it was done. They do not prove that Graph Cut is always better, than Watershed because changing the picture, thresholds, markers or GrabCut start can change the outcome.

Google colab link

4.2 Comparative Performance and Error Analysis

Criterion	**Watershed**	**Graph Cut / GrabCut**
Basic principle	Topographic / marker-based region separation	Foreground-background energy optimization
Initialization	Requires useful foreground/background markers	Rectangle or other initial labeling
Strength	Useful for separating regions with clear boundaries	Effective for foreground isolation with useful initialization
Main issue	Sensitive to markers and possible over-segmentation	Can depend on initialization and color similarity
Observed result	Small/incomplete isolated region in this experiment	Larger connected scene region isolated, according to supplied interpretation
Suitable condition	Strong regional boundaries and reliable markers	Rough foreground region can be indicated

The comparison shows that the two methods use types of information for segmentation. Watershed depends a lot on the markers that come from thresholding and morphology while GrabCut uses an area and keeps updating the model for foreground and background. So the way each method works in practice depends on how its assumptions fit the scene being processed.

The experiment also shows an idea about analyzing errors: a segmentation that doesn't look complete might be because of a choice made earlier in the process not because the segmentation step failed. In the Watershed process choosing the threshold and making good markers has a big effect, on the final foreground mask. In GrabCut the first rectangle and how the image looks in terms of foreground and background affect the result.

4.3 Sensitivity, Interpretability and Practical Observations

Watershed is more understandable because the steps that lead to the result can be looked at one by one: converting to grayscale, smoothing, thresholding, morphology, distance transformation, connected components and markers. This allows me to find where an error happened in a preprocessing step.

GrabCut also gives a workflow at the start because the user can see how the initial rectangle matters. Yet the repeated building of the foreground and background models means that the final pixel labels rely on the image statistics that GrabCut learns while running.

The main practical point from the example is that segmentation must be examined with how it is started. A fair comparison therefore needs the input image to stay the same the preprocessing settings to be written down and a clear note, on whether ground-truth annotations exist or not.

4.4 Challenges and Solutions

- Natural scenes contain several overlapping regions. Solution: apply suitable preprocessing and marker generation before Watershed.
- Thresholding may select unwanted regions. Solution: tune the threshold and morphological parameters for the selected image.
- GrabCut depends on initialization. Solution: use a rectangle that encloses the intended foreground and refine it when necessary.
- Visual comparison does not provide a numerical accuracy score. Solution: use manually annotated ground-truth masks and metrics such as IoU and F1 when quantitative evaluation is required.

5. CONCLUSION AND FUTURE SCOPE

5.1 Summary of Project Achievements

This project successfully. Implemented a comparative image-segmentation workflow for Watershed and Graph Cut/GrabCut using a natural mountain-lake image. The complete workflow covers image loading, resizing when necessary, preprocessing, marker construction, segmentation, mask generation, isolated-object creation and visual output generation.

The Watershed method demonstrated marker-based segmentation using thresholding, morphological operations, distance transformation and connected components. The quality of its isolated-object result was strongly affected by the generated foreground mask. The Graph Cut/GrabCut method used a rectangle and iterative foreground-background modeling and the supplied experiment reported a larger connected foreground region for the selected image and initialization.

The project therefore demonstrates that segmentation performance is dependent on the interaction between the algorithm, image content and initialization or preprocessing choices. The result of one image- experiment should be treated as a case observation rather, than a universal performance ranking.

5.2 Broad Recommendations for Improved Segmentation

Use more reliable foreground and background marker generation when using Watershed on natural scenes.Adjust threshold and morphological parameters based on the traits of the image that is being used.

Use a placed GrabCut initialization rectangle and change it if the correct foreground is not separated.Keep the input settings when comparing segmentation algorithms so that any differences are because of the methods and how they are set up.When it is possible check segmentation results with created masks and numbers that measure performance along, with looking at the images.

5.3 Limitations and Directions for Future Research

The main limitation of the experiment shown is that it uses one natural-scene image. One image alone does not cover the variety of foreground-background situations that happen in real-world computer vision tasks. Also the Watershed method relies on threshold values and marker choices while GrabCut depends on the rectangle and the images features.

Another issue is that there is no labeled ground-truth segmentation mask. Because of this the experiment did not include evaluation metrics like pixel accuracy Intersection over Union (IoU) precision, recall or F1 score.

Future work could include images with different object and background setups. Creating ground-truth masks would allow for quantitative testing. The Watershed markers could be improved by using foreground and background detection methods. GrabCut could also be tested with initializations. Other techniques, like k-means clustering, region growing and learning-based segmentation could be added to make the comparison more complete and meaningful.

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